DECISION ML COURSE PROJECT · HUMANOIDVERSE / GENESIS / PPO

Stable Arm-Swing Humanoid Locomotion under Upper-Body Perturbations

A staged reward-routing curriculum preserves full-body walking, introduces coordinated shoulder-pitch arm swing, and evaluates recovery under torso, shoulder, and elbow action pulses.

Course Project Team · H1 Humanoid · 19DoF full-body control · Headless evaluation in Genesis

Robust walking is learned without freezing upper-body motion.

The final policy survives strong upper-body perturbations while retaining visible shoulder swing and forward command tracking.

19	CONTROLLED DOF
32	PARALLEL ENVS
20s	EVAL HORIZON
0	FALLS AT AMP2

Dynamic Evidence: from Locked Upper Body to Perturbation Recovery

FIGURE 1 · TEMPORAL ROLLOUTS

Problem formulation

Once the upper body is unlocked, humanoid locomotion is no longer only a lower-body tracking problem. Torso, shoulder, and elbow actions perturb angular momentum, contact timing, and posture, so a naive arm-swing reward can degrade the base gait.

Contributions

- Full-body H1 control in a 19DoF action space.
- Command-routed reward families that protect walking before arm shaping.
- Upper-body action-pulse evaluation on torso, shoulder, and elbow channels.
- Qualitative temporal evidence paired with finite-horizon metrics.

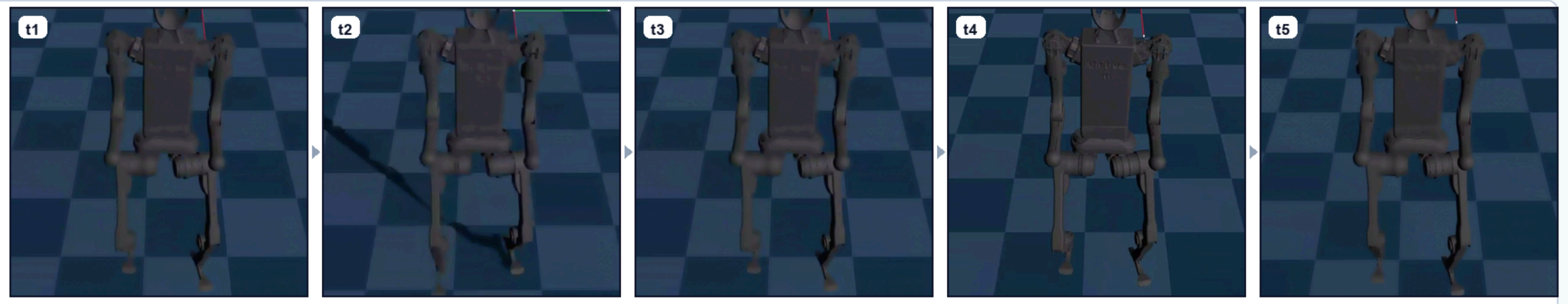
Task-to-benchmark mapping

Forward locomotion	0.6 m/s command tracking under stable gait.
Upper body motion	Visible shoulder-pitch swing synchronized with walking.
Robustness	Recovery after held upper-body action offsets.

Design principle

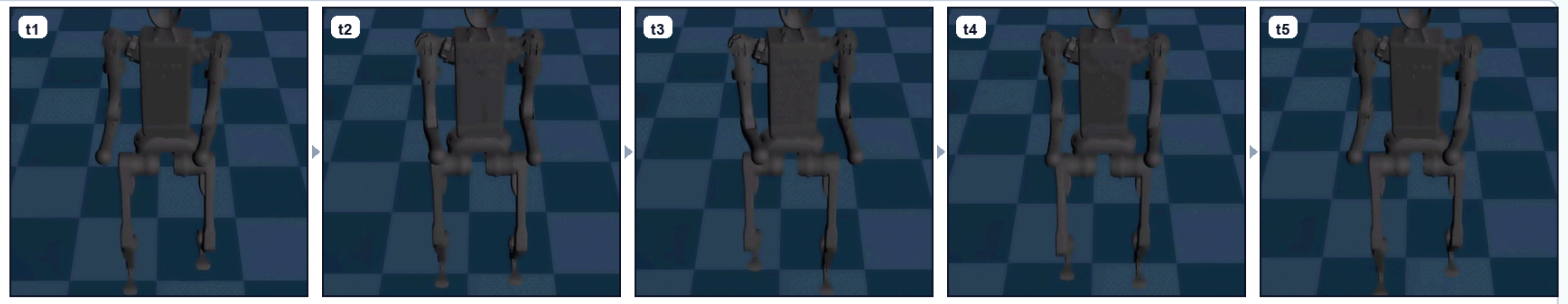
Locomotion is protected before expressiveness is added. Command-derived masks activate standing, walking, and straight-walk reward terms only in compatible regimes.

A.
Locked upper-body baseline
stable gait, upper body constrained



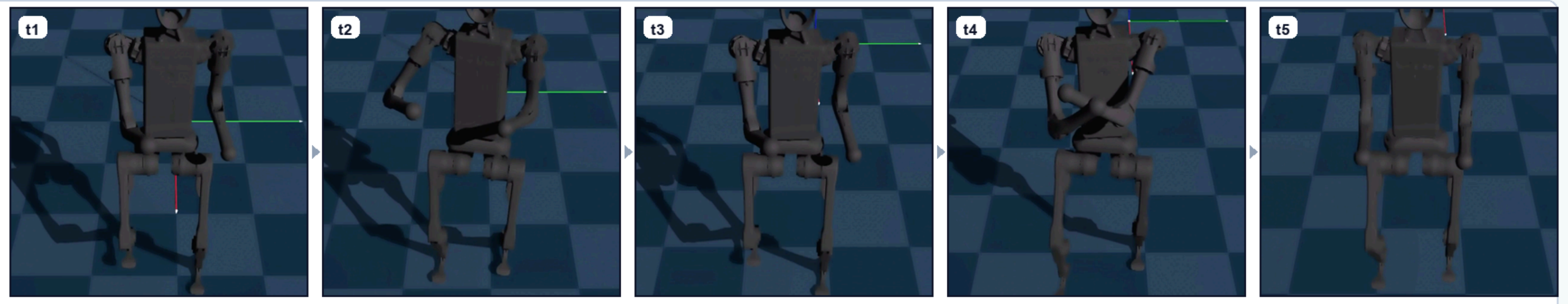
The policy first preserves stable forward walking with the upper body held near default posture.

B.
Coordinated arm swing
visible shoulder-pitch swing



After arm-swing shaping, shoulder motion becomes visible while the walking rhythm is maintained.

C.
Upper-body perturbation recovery
pulse disturbance and recovery



Upper-body action offsets disturb torso and arms; the policy continues walking after the pulse window.

Evaluation protocol

We evaluate 32 parallel H1 humanoids for a 20 s horizon. Upper-body action pulses are applied to torso, shoulder, and elbow channels while the forward command remains fixed at 0.6 m/s.

Metrics reported

Survival counts environments without non-timeout resets. Tracking error measures forward velocity deviation. Max tilt uses the projected-gravity tilt proxy and exposes perturbation-induced posture excursions.

Result interpretation

Recovery success is not used as the headline baseline comparison because a policy may briefly satisfy the recovery criterion after some pulses while still repeatedly resetting. Survival, falls, and max tilt are more decisive.

Reading the rows

Row A verifies the stable walking foundation. Row B shows visible gait-synchronized arm swing. Row C shows a perturbed upper body returning to forward walking.

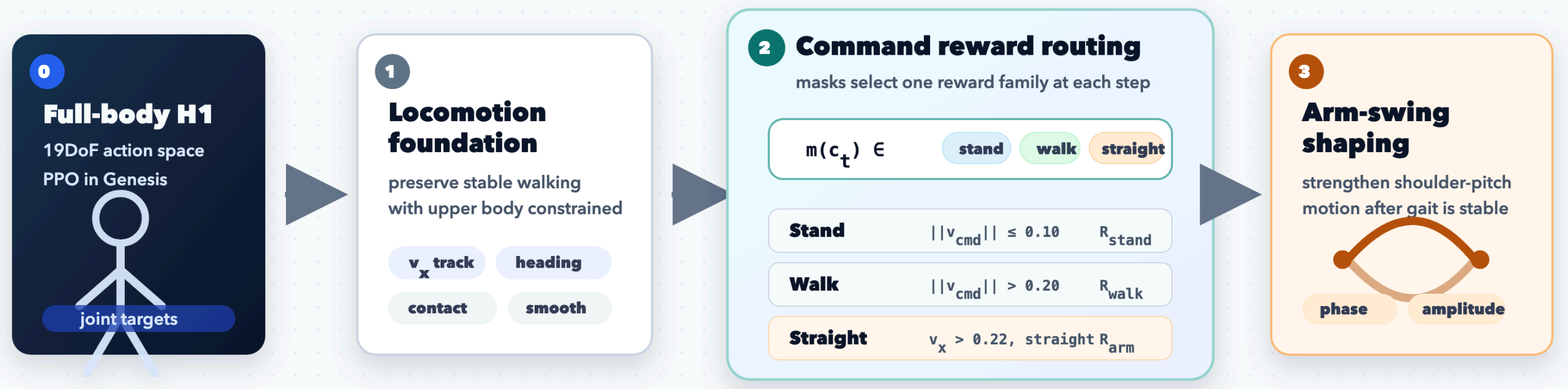
Temporal strips compare stable locked-upper-body walking, gait-synchronized arm swing, and recovery after an injected upper-body action pulse. The qualitative claim is temporal: the robot continues walking while upper-body behavior changes from constrained, to coordinated, to externally perturbed.

Method: Staged Reward Routing

FIGURE 2 · METHOD

Staged Reward Routing for Full-Body Arm-Swing Locomotion

Reward terms are activated by command regimes so arm-swing objectives do not destabilize the base gait.



0 **Full-body H1**
19DoF action space
PPO in Genesis
joint targets

1 **Locomotion foundation**
preserve stable walking with upper body constrained
v_x track · heading · contact · smooth

2 **Command reward routing**
masks select one reward family at each step
$m(c_t) \in$ stand · walk · straight

Stand	$\|\|v_{cmd}\|\| \le 0.10$	R_{stand}
Walk	$\|\|v_{cmd}\|\| > 0.20$	R_{walk}
Straight	$v_x > 0.22$, straight	R_{arm}

3 **Arm-swing shaping**
strengthen shoulder-pitch motion after gait is stable
phase · amplitude

Routed objective

$$R_t = R_{shared} + m_{stand}R_{stand} + m_{walk}R_{walk} + m_{straight}R_{arm} - C_{upper}$$

● shared rewards stay active: velocity, heading, contact, smoothness
● arm-swing terms activate mainly during straight walking

The staged protocol preserves the locomotion solution before strengthening upper-body expressiveness.
Robustness is reported separately in the perturbation results, not mixed into the reward-routing pipeline.

The controller is trained in the full 19DoF H1 action space. Stable locomotion is protected first; command masks then activate stand, walk, and straight-walk reward families so arm-swing objectives appear mainly when they are compatible with forward walking.

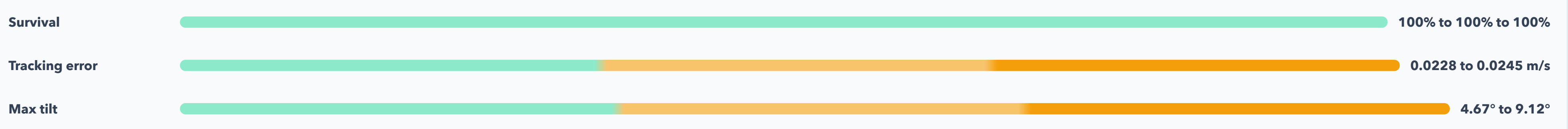
Quantitative Robustness under Upper-Body Pulses

100%
survival at amp = 2.0 across 32 environments

0 vs. 95
falls: robust policy vs. arm-swing baseline under amp2

POLICY / SETTING	AMP	SURVIVAL	MEAN TIME	FALLS	TRACK ERR.	MAX TILT
Robust policy	0.0	100%	20.00s	0	0.0228	4.67°
Robust policy	1.0	100%	20.00s	0	0.0225	7.81°
Robust policy	2.0	100%	20.00s	0	0.0245	9.12°
Arm-swing baseline	2.0	0%	5.48s	95	0.1401	89.93°

Robust policy under increasing perturbation amplitude



The amp0 run uses the same perturbation scheduling with zero magnitude. At amp2, the robust policy retains arm motion: shoulder-pitch swing amplitude 0.2536 rad and elbow sagittal separation 0.1165 m, so robustness is not obtained by simply freezing the upper body.

Experimental Story

1	**Locomotion foundation** Full-body H1 walks forward while the upper body is constrained to preserve a reliable gait.
2	**Command-routed preparation** Stand, walk, and straight-walk masks reduce interference between posture, gait, and arm objectives.
3	**Coordinated arm swing** Shoulder-pitch phase and endpoint-sagittal terms make the upper body visibly participate in walking.
R	**Perturbation-robust policy** Held action offsets on torso, shoulder, and elbow channels test whether the gait recovers.

Takeaway: unlocking upper-body joints is not enough. Stable arm-swing locomotion emerges when walking is protected first, arm coordination is routed by command regime, and robustness is evaluated with explicit upper-body disturbances.

Limitations

Evidence is simulation-only and short-horizon. Upper-body naturalness is shaped by hand-designed phase and amplitude rewards rather than human motion priors. Future work should add angular-momentum analysis, energy cost, longer horizons, and cleaner camera evaluation.

Conclusion

The final controller demonstrates stable forward walking, visible arm swing, and bounded recovery behavior under explicit upper-body disturbances, forming a compact paper-style story from curriculum design to quantitative robustness.

Simulation-only course evaluation. Key references and context: HumanoidVerse, Genesis, PPO-based humanoid locomotion, gait-conditioned reward routing, Humanoid-Gym, DeepMimic, and adversarial/motion-prior humanoid control literature. Quantitative values are finite-horizon server evaluations, not sim-to-real claims.

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